**9** Car *A* and car *B* take part in a race around a circular track.
One lap of the track measures 7.6 km.

Car *A* takes 2 minutes and 40 seconds to complete each lap of the track.
Car *B* takes 2 minutes and 25 seconds to complete each lap of the track.
Both cars travel at a constant speed.

**(a)** Calculate the speed of car *A*.
Give your answer in kilometres per hour.

..... km/h [3]

**(b)** Both cars start the race from the same position, *S*, at the same time.

**(i)** Find the time taken when both car *A* and car *B* are next at position *S* **at the same time**.
Give your answer in minutes and seconds.

..... min ..... s [4]

**(ii)** Find the distance that car *A* has travelled at this time.

..... km [2]